

# CS & IT ENGINEERING



## Computer Network

### Transport Layer

**Lecture No. - 05**



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# Recap of Previous Lecture



Topic

TCP Connections

Topic

TCP Data transfer



# Topics to be Covered



Topic

TCP Data transfer

# ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. Which of the following statements are TRUE for 'three way handshake' for TCP connection establishment?

- (S1) Loss of (SYN + ACK) from the server will not establish a connection TRUE
- (S2) Loss of ACK from the client cannot establish the connection FALSE
- (S3) The server moves LISTEN → SYN\_RCVD → SYN\_SENT → ESTABLISHED in the state machine on no packet loss → FALSE
- (S4) The server moves LISTEN → SYN\_RCVD → ESTABLISHED in the state machine on no packet loss. TRUE

[GATE-2008]

- ~~(A) S2 and S3 only~~
- ☒ (B) S1 and S4 only
- ~~(C) S1 and S3 only~~
- (D) S2 and S4 only

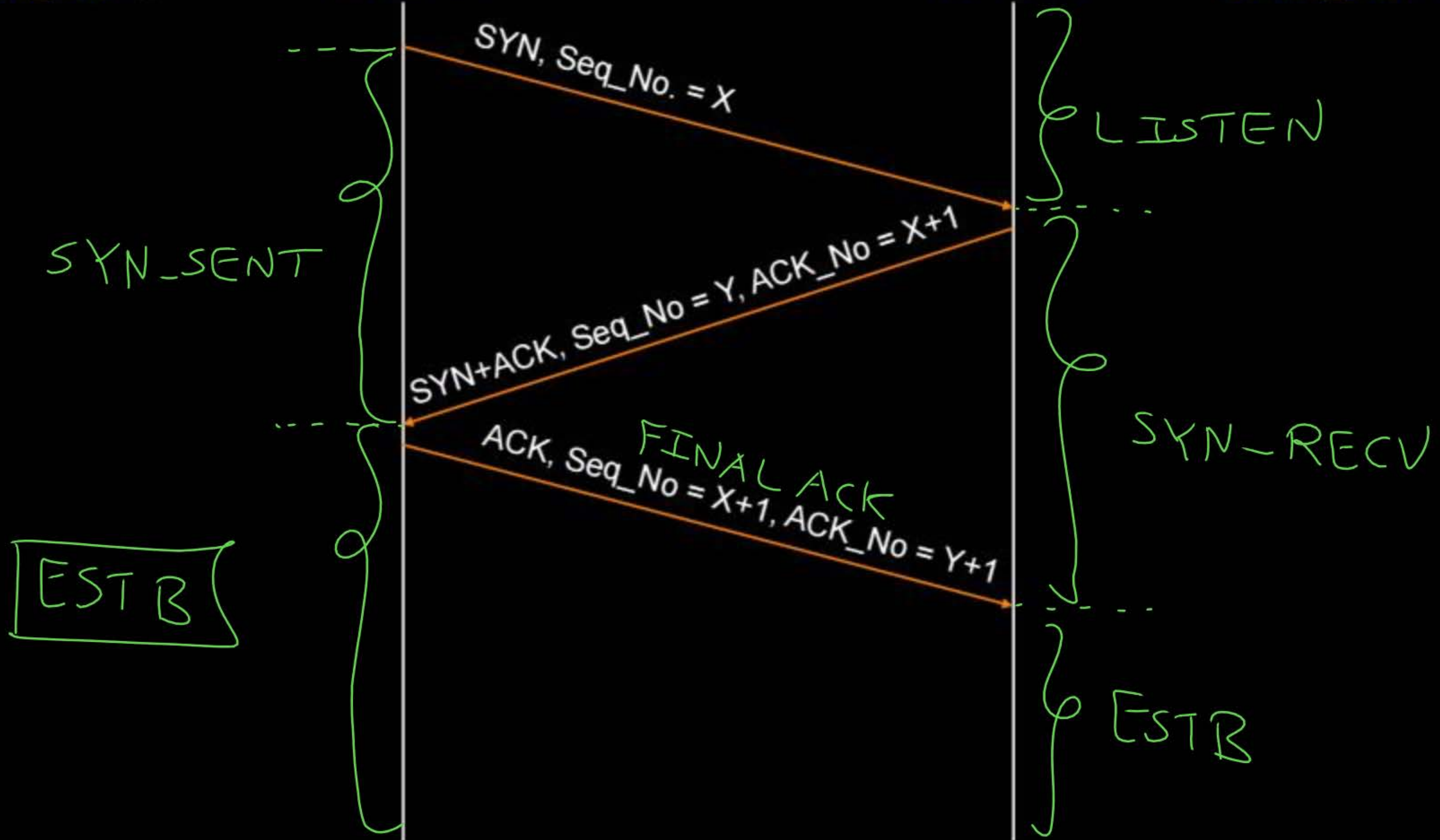
Ans: B

Client\_ISN = X

TCP Client

TCP Server

Server\_ISN = Y



TCP Client Process

$P_1$

SOCK\_STREAM  
(TCP Socket)

LastByteWritten =  $P_1$ 's ISN = ~~299~~/899

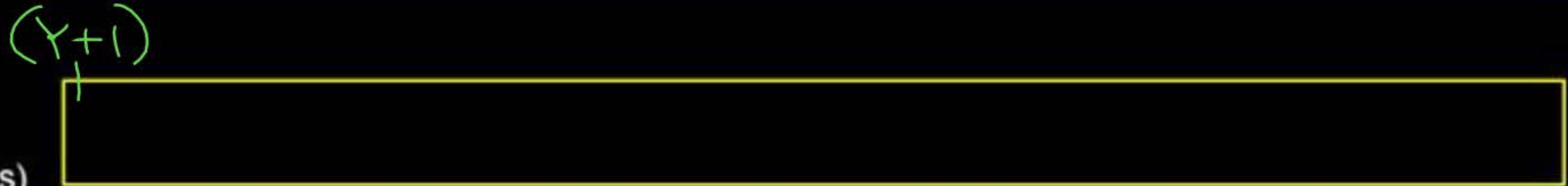
LastByteSent =  $P_1$ 's ISN = ~~299~~/~~399~~/~~499~~/599

LastByteAcked =  $P_1$ 's ISN = ~~299~~/~~399~~/~~499~~/599

Send Buffer  
(MaxSendBuffer = 1000 Bytes)



Recv Buffer  
(MaxRecvBuffer = 1000 Bytes)



LastByteRead =  $P_2$ 's ISN =  $Y$

NextByteExpected =  $P_2$ 's ISN + 1 =  $Y+1$

LastByteRcvd =  $P_2$ 's ISN =  $Y$



## Topic : TCP



=> **Maximum Segment Size (MSS)** : Maximum bytes can be kept in the payload field of a TCP segment

=> TCP protocol wants to transmit next segment from process  $P_1$  'Send Buffer'

→ Fill upto MSS bytes [ (LastByteSent + 1) to (LastByteSent + MSS) ]  
in the payload field of segment from 'Send Buffer'

→ Sequence Number = [ LastByteSent + 1 ];

→ ACK Number = NextByteExpected;

→ LastByteSent = LastByteSent + MSS;



# Topic : TCP

P<sub>1</sub>

TCP Client

TCP Server

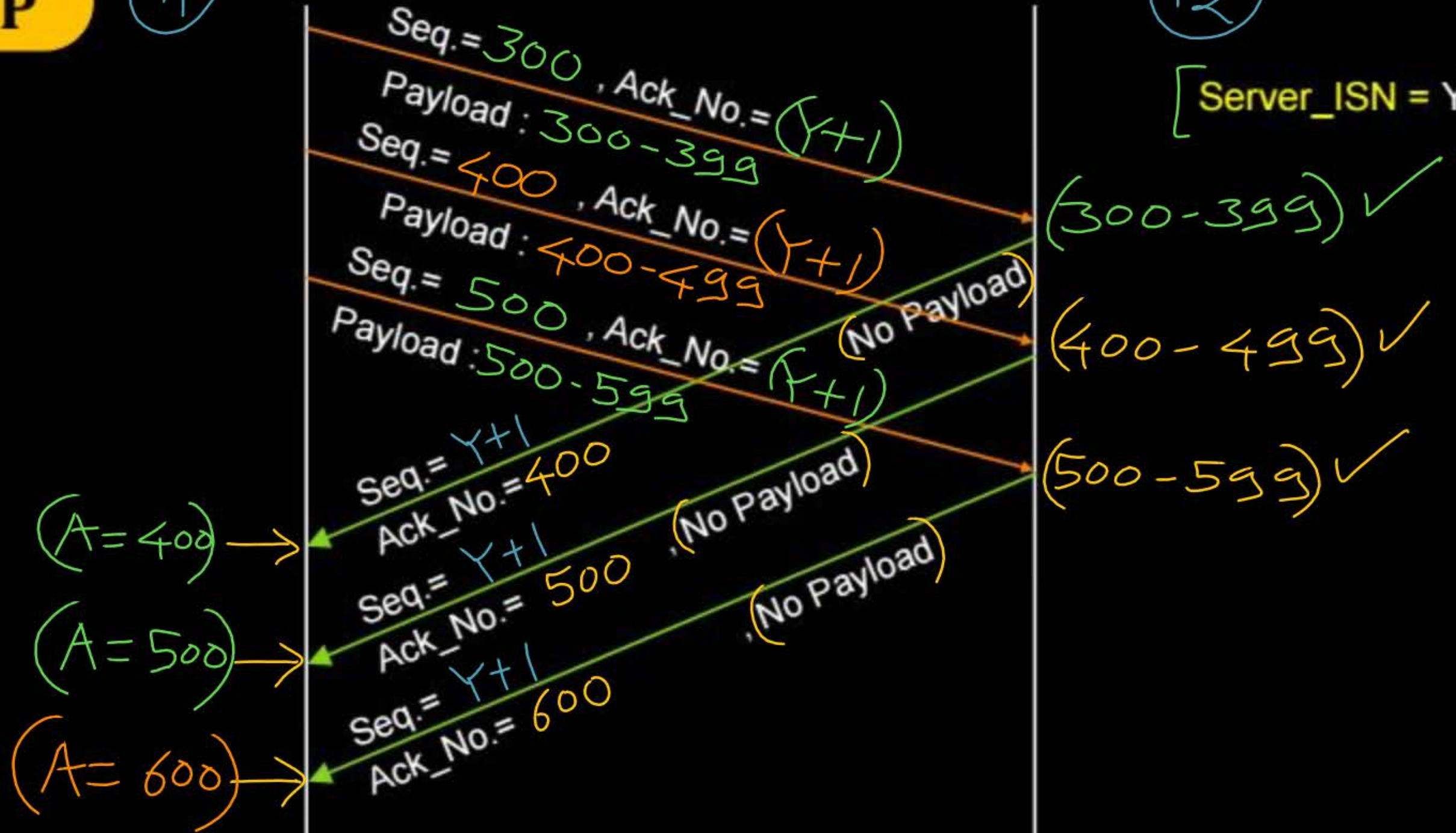
P<sub>2</sub>



[Client\_ISN = 299]

MSS = 100 bytes

[Server\_ISN = Y]





## Topic : TCP



=> The TCP protocol receive a segment from process P<sub>2</sub>,  
contains acknowledgement number A for process P<sub>1</sub>

→ LastByteAcked =  $[A - 1]$ ;

→ Remove all the bytes of P<sub>1</sub> 'Send Buffer' from begin to LastByteAcked



## Topic : TCP



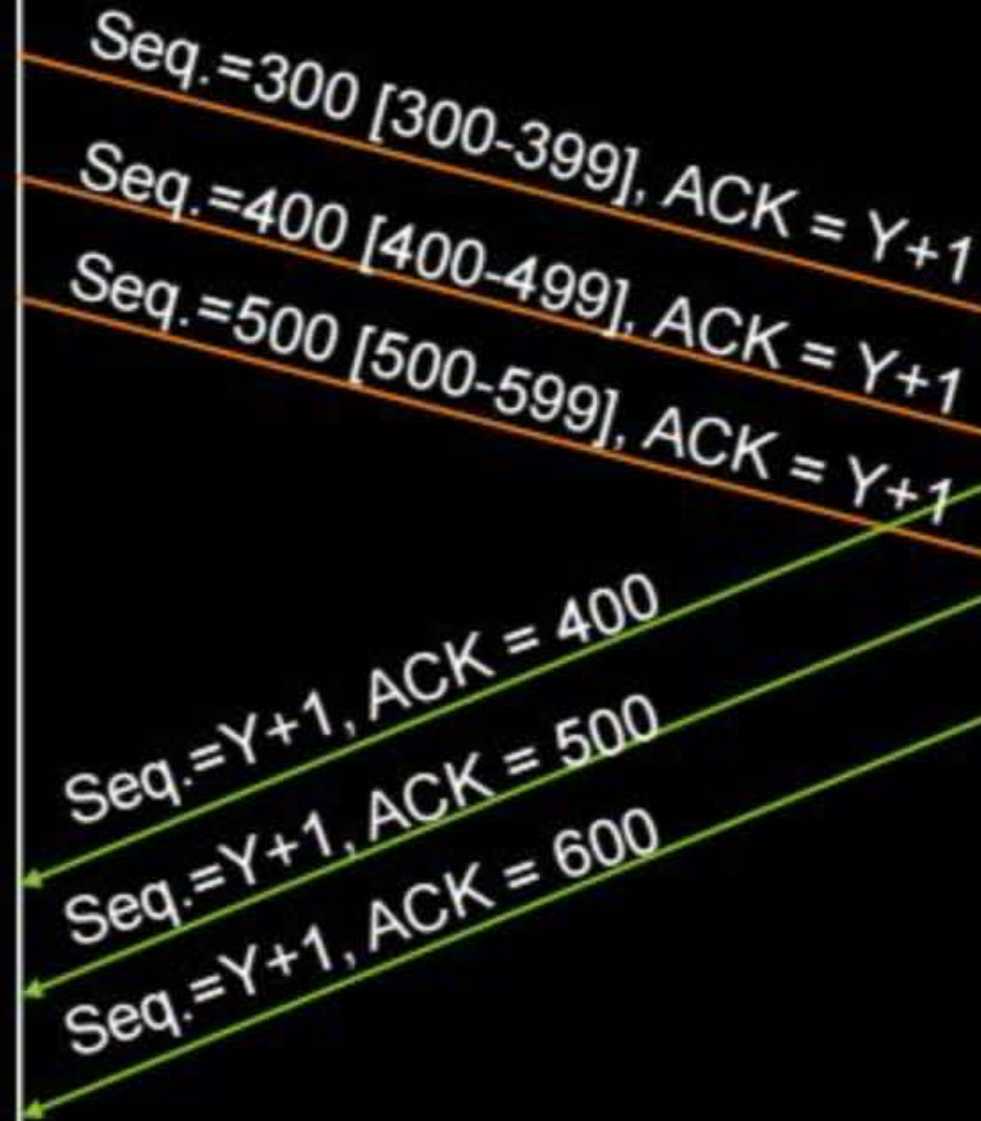
Client\_ISN = 299

MSS = 100 bytes

TCP Client

TCP Server

Server\_ISN = Y





## Topic : Sequence Number



=> TCP is byte-stream protocol (Stream-oriented protocol)

=> Every byte in 'service data unit' is identified uniquely  
[with 32-bit sequence number]

=> Unique sequence number is assigned to each byte ✓

Sequence Number in TCP Segment ← First Byte Sequence Number of Payload

#Q. In TCP, a unique sequence number is assigned to each :

[GATE-2004]

✓ (A) byte

(B) word

~~(C)~~ segment

(D) message

11T-1

Ans: A

# TCP Server Process

$P_2$

SOCK\_STREAM  
(TCP Connection  
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

$(Y+1)$

$(x+1)=300$  400 500

399 499 599

LastByteWritten =  $P_2$ 's ISN =  $Y$

LastByteSent =  $P_2$ 's ISN =  $Y$

LastByteAcked =  $P_2$ 's ISN =  $Y$

LastByteRead =  $P_1$ 's ISN =  $x = 299$

NextByteExpected =  $P_1$ 's ISN + 1 =  $(x+1) = \cancel{300} = \cancel{400} = \cancel{500} = 600$

LastByteRcvd =  $P_1$ 's ISN =  $x = \cancel{299} = \cancel{399} = \cancel{499} = 599$



# Topic : TCP



Client\_ISN = 299

MSS = 100 bytes

TCP Client

TCP Server

Server\_ISN = Y

Seq. = 300 [300-399], ACK = Y+1

Seq. = 400 [400-499], ACK = Y+1

Seq. = 500 [500-599], ACK = Y+1

(Seq. = Y+1, ACK = 400)

(Seq. = Y+1, ACK = 500)

(Seq. = Y+1, ACK = 600)

[300-399] ✓

[400-499] ✓

[500-599] ✓



## Topic : TCP



$$[K = 100]$$

=> The TCP protocol receive a segment (contains K bytes in its payload)  
from process  $P_1$ , for process  $P_2$

- Store payload bytes in process  $P_2$  'Recv Buffer'
- Update variables according to case

=> Send Acknowledgement to sender TCP

$$\begin{aligned}\text{Sequence Number} &= \text{LastByteSent} + 1; \\ \text{ACK Number} &= \text{NextByteExpected};\end{aligned}$$



### CASE I :

if (Sequence Number = NextByteExpected) and  
(LastByteRcvd + 1 = NextByteExpected)

→ Keep payload bytes in process P<sub>2</sub> 'Recv Buffer'

→ NextByteExpected = NextByteExpected + K;

→ LastByteRcvd = LastByteRcvd + K;



### CASE II :

if Sequence Number < NextByteExpected

→ ignore received TCP segment (duplicate segment)

### CASE III :

if (Sequence Number > NextByteExpected)

→ Keep payload bytes in process P<sub>2</sub> 'Recv Buffer'  
after [Sequence Number - NextByteExpected] empty bytes

→ LastByteRcvd = Sequence Number + K - 1;



### CASE IV :

if (Sequence Number = NextByteExpected) and  
(LastByteRcvd > NextByteExpected) and  
(Missing bytes > K)

→ NextByteExpected = NextByteExpected + K;

### CASE V :

if (Sequence Number = NextByteExpected) and  
(LastByteRcvd > NextByteExpected) and  
(Missing bytes = K)

→ NextByteExpected = LastByteRcvd + 1;

# TCP Server Process

$P_2$

SOCK\_STREAM  
(TCP Connection  
Socket 1)

Send Buffer

(MaxSendBuffer = 1200 Bytes)

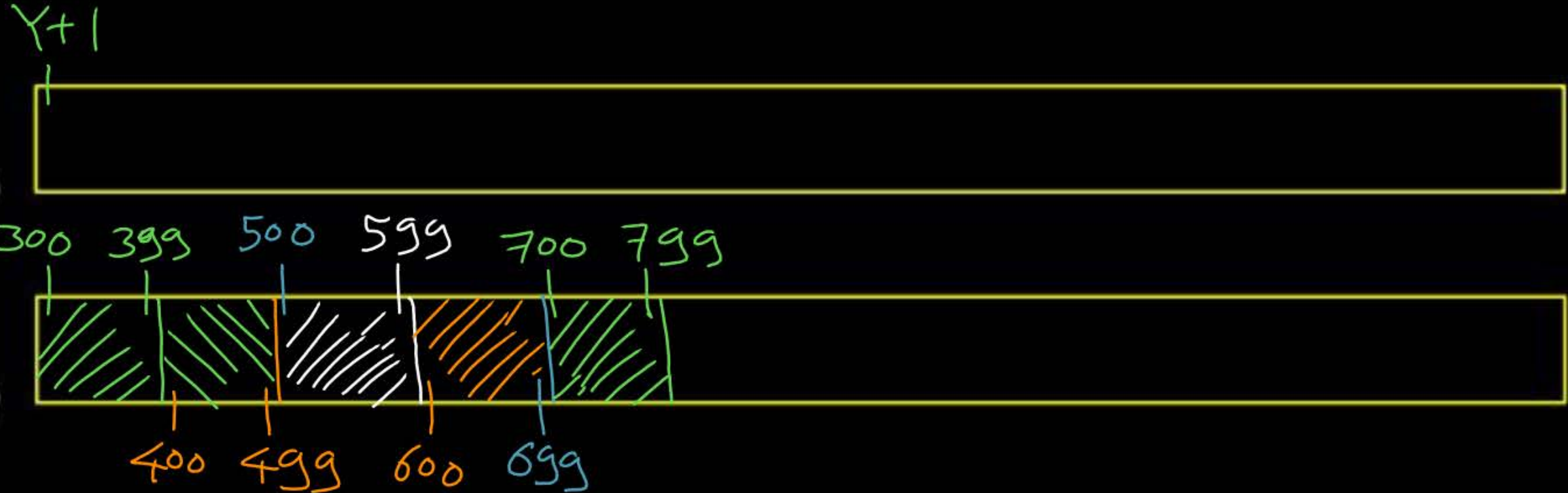
Recv Buffer

(MaxRecvBuffer = 1200 Bytes)

LastByteWritten =  $P_2$ 's ISN =  $Y$

LastByteSent =  $P_2$ 's ISN =  $Y$

LastByteAcked =  $P_2$ 's ISN =  $Y$



LastByteRead =  $P_1$ 's ISN = 299

NextByteExpected =  $P_1$ 's ISN + 1 = ~~300~~ = ~~400~~ / ~~500~~ / ~~600~~ / ~~800~~

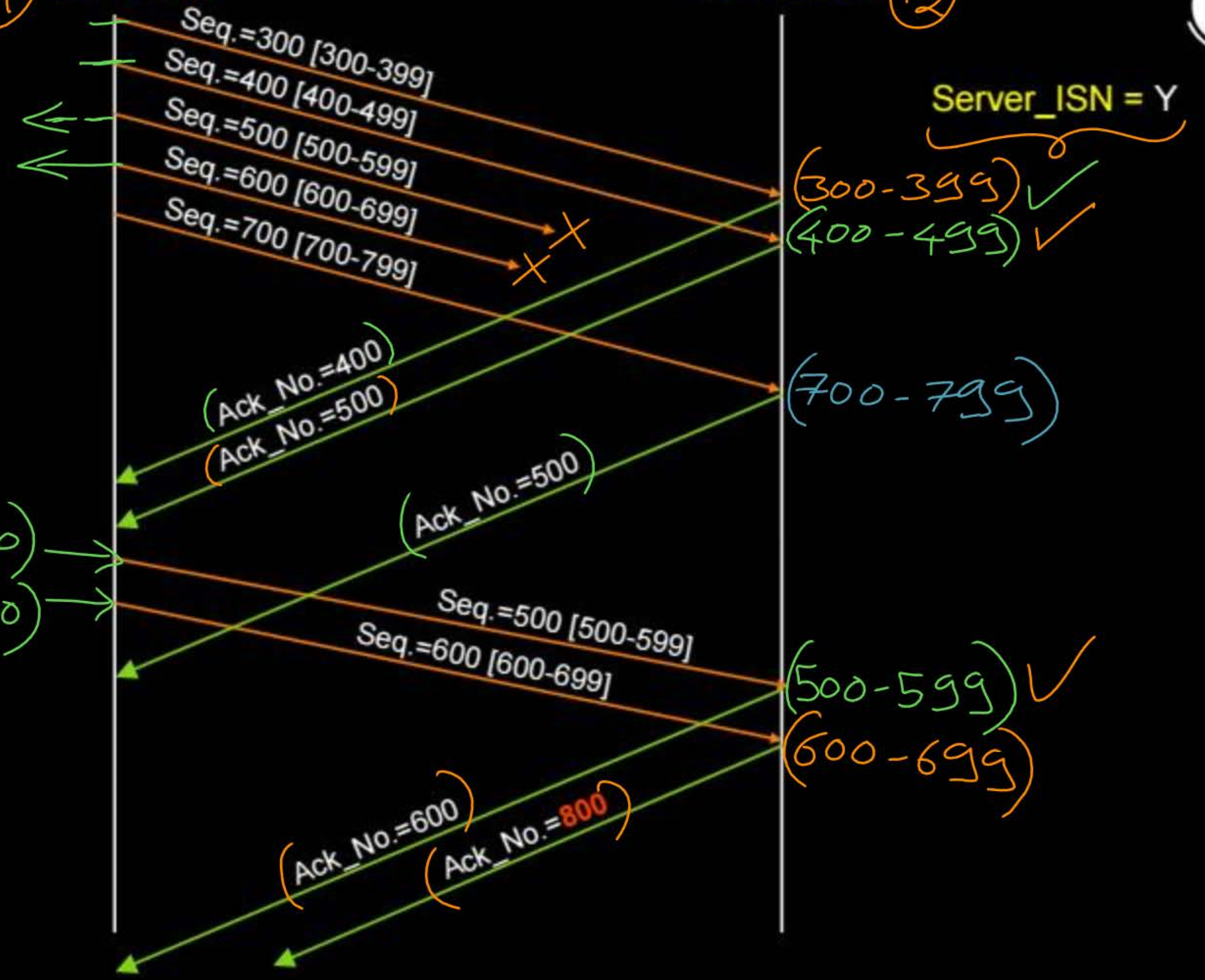
LastByteRcvd =  $P_1$ 's ISN = ~~299~~ = ~~399~~ / ~~499~~ / ~~799~~



# Topic : TCP

[ Client\_ISN = 299 ]  
[ MSS = 100 bytes ]

Re To (500)  
Re To (600)





## Topic : TCP



→ The process  $P_2$  read  $N$  bytes from 'Recv Buffer'

→  $\text{LastByteRead} = \text{LastByteRead} + N$  ;

→ Remove  $N$  bytes from  $P_2$  'Recv Buffer' from begin



## 2 mins Summary



Topic

TCP Data transfer



# THANK - YOU

